Practical No 01

a) Develop a simple "Hello World" Android application demonstrating project creation, Android Studio interface, Android components, AVD creation, and application execution.

Project Name: HelloWorld

Template: Empty Views Activity

Language: Java

Program Code:

I. Filename: MainActivity.java

```
package com.example.helloworld;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
  }
}
```

II. Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <TextView
        android:id="@+id/text_hello"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="6137"
        android:textSize="24sp"
        android:textStyle="bold"
        android:textColor="#000000" />

</LinearLayout>
```

III. Filename: AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.helloworld">

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Helloworld">

        <activity android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
```

```
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

IV. Filename: colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="purple_500">#6200EE</color>
    <color name="purple_700">#3700B3</color>
</resources>
```

V. Filename: styles.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <style name="Theme.Helloworld"
parent="Theme.MaterialComponents.DayNight.DarkActionBar">
        <item name="colorPrimary">@color/purple_500</item>
        <item name="colorPrimaryVariant">@color/purple_700</item>
        <item name="colorOnPrimary">@color/white</item>
    </style>
</resources>
```

b) Develop an Android application demonstrating the use of Programming Resources (colors.xml, themes.xml, strings.xml, dimens.xml, drawable resources) and apply them in UI components.

Project Name: Practical2R

Template: Empty Views Activity

Language: Java

Program Code:

I. Filename: AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light"
        tools:targetApi="31">

        <activity android:name=".ThemeActivity" android:label="Theme Demo" />
        <activity android:name=".StringActivity" android:label="String Demo" />
        <activity android:name=".DrawableActivity" android:label="Drawable Demo" />
        <activity android:name=".DimensionActivity" android:label="Dimension Demo"
/>

        <activity
            android:name=".MainActivity"
            android:exported="true"
            android:label="@string/app_name"
            android:theme="@style/Theme.AppCompat.Light">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

II. Filename: MainActivity.java

```
package com.example.practical2r;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    public void openThemeActivity(View view) {
        startActivity(new Intent(this, ThemeActivity.class));
    }
```

```
    public void openStringActivity(View view) {
        startActivity(new Intent(this, StringActivity.class));
    }
    public void openDrawableActivity(View view) {
        startActivity(new Intent(this, DrawableActivity.class));
    }
    public void openDimensionActivity(View view) {
        startActivity(new Intent(this, DimensionActivity.class));
    }
}
```

III. Filename: ColorActivity.java

```
package com.example.practical2r;

 import android.os.Bundle;
 import androidx.appcompat.app.AppCompatActivity;

 public class ColorActivity extends AppCompatActivity {
      @Override
      protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_color);
      }
 }
```

IV. Filename: ThemeActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class ThemeActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_theme);
    }
}
```

V. Filename: StringActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class StringActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_string);
  }
}
```

VI. Filename: DrawableActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class DrawableActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_drawable);
  }
}
```

VII. Filename: DimensionActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class DimensionActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_dimension);
  }
}
```

VIII. Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"
    android:background="@drawable/pebbles">

    <Button
        android:id="@+id/btnTheme"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Theme Demo"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="50dp"
        android:onClick="openThemeActivity" />

    <Button
        android:id="@+id/btnString"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="String Demo"
        android:layout_below="@id/btnTheme"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"
        android:onClick="openStringActivity" />

    <Button
        android:id="@+id/btnDrawable"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:text="Drawable Demo"
        android:layout_below="@id/btnString"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"
        android:onClick="openDrawableActivity" />

    <Button
        android:id="@+id/btnDimension"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Dimension Demo"
        android:layout_below="@id/btnDrawable"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"
        android:onClick="openDimensionActivity" />

</RelativeLayout>
```

IX. Filename: activity_color.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:background="@color/colorRed">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Color Activity"
      android:textSize="24sp"
          android:layout_centerInParent="true"
      android:textColor="@color/white"/>
 </RelativeLayout>
```

X. Filename: activity_dimension.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:orientation="vertical"
      android:padding="16dp"
      android:gravity="center">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Dimension Activity"
      android:textSize="24sp"
      android:textColor="@android:color/black"/>
      <!-- Button demonstrating dimension usage -->
      <Button
```

```
        android:id="@+id/dimensionButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Click Me!"
    android:textSize="@dimen/button_text_size"
    android:padding="@dimen/button_padding"
    android:layout_marginTop="@dimen/button_margin_top"
    android:background="@color/btn_color"/>
</LinearLayout>
```

XI. Filename: activity_theme.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@android:color/holo_blue_bright">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="You got this sky blue theme"
        android:textSize="24sp"
        android:layout_centerInParent="true"
        android:textColor="@android:color/black" />

</RelativeLayout>
```

XII. Filename: activity_string.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:background="@color/colorPrimaryDark">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/Welcome_Message"
        android:textSize="24sp"
        android:textColor="@android:color/white" />

</LinearLayout>
```

XIII. Filename: activity_drawable.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/pebbles">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Drawable Activity"
```

```
        android:textSize="24sp"
        android:layout_centerInParent="true"
        android:textColor="@android:color/white" />

</RelativeLayout>
```

XIIV. Filename: activity_dimension.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Dimension Activity"
        android:textSize="24sp"
        android:textColor="@android:color/black" />

    <Button
        android:id="@+id/dimensionButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click Me!"
        android:textSize="@dimen/button_text_size"
        android:padding="@dimen/button_padding"
        android:layout_marginTop="@dimen/button_margin_top"
        android:background="@color/btn_color" />

</LinearLayout>
```

XV. Filename: colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="purple_200">#FFBB86FC</color>
    <color name="purple_500">#FF6200EE</color>
    <color name="purple_700">#FF3700B3</color>
    <color name="teal_200">#9ADA03</color>
    <color name="teal_700">#FF018786</color>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="btn_color">#FFC49662</color>
    <color name="colorPrimary">#6200EE</color>
    <color name="colorPrimaryDark">#3700B3</color>
    <color name="colorAccent">#03DAC5</color>
</resources>
```

XVI. Filename: dimens.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <dimen name="button_margin_top">24dp</dimen>
    <dimen name="button_padding">16dp</dimen>
    <dimen name="button_text_size">18sp</dimen>
</resources>
```

XVII. Filename: strings.xml

```
<resources>
    <string name="app_name">Practical2R</string>
    <string name="ParaHead">Welcome to Practical 2R</string>
    <string name="Welcome_Message">6137</string>
</resources>
```

XVIII. Filename: themes.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <style name="AppTheme" parent="Theme.AppCompat.Light.DarkActionBar">
        <item name="colorPrimary">@color/colorPrimary</item>
        <item name="colorPrimaryDark">@color/colorPrimaryDark</item>
        <item name="colorAccent">@color/colorAccent</item>
    </style>
</resources>
```

c) Develop an Android application demonstrating complete Activity Lifecycle methods and analyse execution using Logcat.

Project Name: LogCatDemo
Template: Empty Views Activity
Language: Java

Program Code:

I. Filename: MainActivity.java

```
package com.example.logact;

import android.os.Bundle;
import android.util.Log;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override
    protected void onStart() {
        super.onStart();
        Log.d("on start invoked", "on started");
    }

    @Override
    protected void onPostResume() {
        super.onPostResume();
        Log.d("on resume invoked", "on resume");
    }

    @Override
    protected void onPause() {
        super.onPause();
        Log.d("on pause invoked", "on pause");
    }

    @Override
    protected void onStop() {
        super.onStop();
        Log.d("on stop invoked", "on stopped");
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d("on destroy invoked", "on destroy");
    }
}
```

II. Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center">
```

```
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Activity Lifecycle Demo"
        android:textSize="20sp" />

</LinearLayout>
```

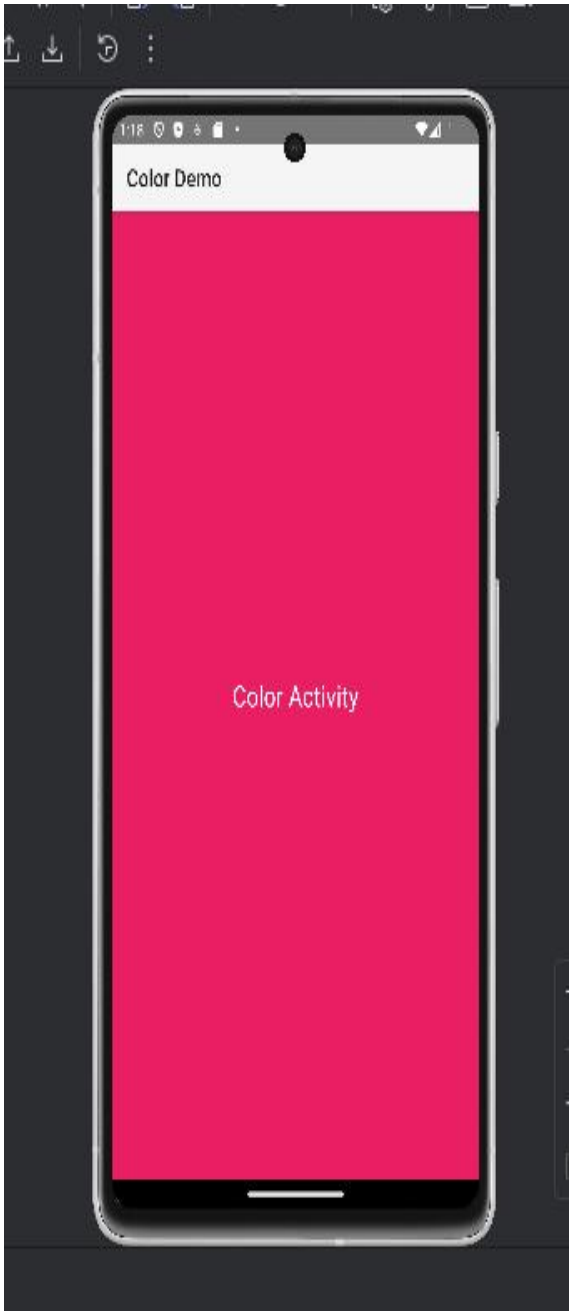
a) **Output:**



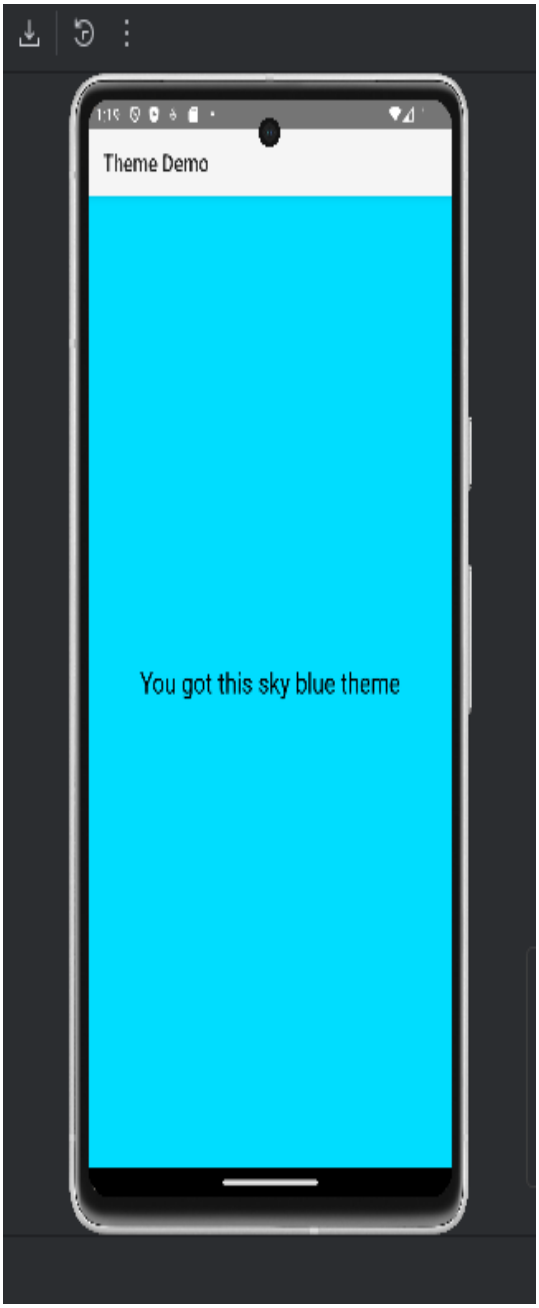
b) Output:



Main Demo



Color Demo



Theme Demo



String Demo



Drawable Demo



Dimension Demo

c) Output: (Logcat)

